



a conventional non-isolated boost converter is proposed to convert the hybrid supercapacitor voltage of 2.5-3.5V to a regulated 5V output. The MOSFET stores energy in L1 during its ON interval and during its OFF interval, L1 transfers energy through D1 to C1 and the load R1

$$V_{out} = \frac{V_{in}}{1-D}$$

$$D = 1 - \frac{V_{in}}{V_{out}}$$

$$D(2.5V) = 0.5, \quad D(3.5V) = 0.3$$

$$R_{load} = \frac{V_{out}^2}{P_{out}} = 5 - 50 \Omega$$

$$L_{min} = \max \left[\frac{2 R_{load} D (1-D)^2}{2 f_s} \right] \approx 37 \mu H$$

$$L = 47 \mu H$$

$$\bullet \text{ At } V_{in} = 2.5V, P_{out} = 5W$$

$$I_{out} = 1A, I_{in} = 2A$$

$$\Delta I_L = \frac{V_{in} D}{L f_s} = 0.266A$$

$$I_{L, peak} = I_{in} + \frac{\Delta I_L}{2} = 2.13A$$

$$C_{o, min} = \frac{I_{out} D}{f_s \Delta V_{out}} = 100 \mu F$$

$$\bullet C_o = 220 \mu F$$

$$\Delta V_{out, ideal} = \frac{I_{out} D}{f_s L_o} = 22.7mV < 50mV$$

Ideal PLECS simulation verifies the topology and initial component values. Closed loop UC3843/ ATmega control and non-ideal device losses will be incorporated during detailed design